

7th International Conference on Data Science and Machine Learning Applications (CDMA2022)

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Abstract—Welcome to the seventh edition of the biennial International Conference on Data Science and Machine Learning Applications (CDMA2022). This edition celebrates the 2030 Saudi Vision and its ambitions goals to reach the top-five nations in AI advancements and deployments.

Keywords—Data Science/Analytics, Machine Learning, Deep Learning.

I. INTRODUCTION

The current edition of the International Conference on Data Science and Machine Learning Applications (CDMA2022) offers a unique opportunity and exchange platform to academic researchers and industry experts from a wide range of Data Science and Machine Learning areas including, but not restricted to, data analytics, computational intelligence, deep learning, pattern recognition, computer vision, multimedia/intelligent databases, big data, visualization, and related ethical issues. The networking and collaboration opportunities are availed to data science and machine learning researchers, developers and practitioners in Saudi Arabia, countries of the Gulf Cooperation Council (GCC), the Middle East and the global community.

The first six editions of the CDMA Conference have been organized an international symposium. The 2020 edition (CDMA2020) marked the start of the international conference format thanks to the success of the past editions. [1]. Like the previous edition, this year conference is technically co-sponsored by the IEEE Region 8 and the IEEE Saudi Arabia Section.

World-class guest speakers and presenters are taking part in the CDMA2022 Conference. A rich and diversified technical program is proposed. The technical program includes 3 different tracks (machine learning, deep learning, and applications), two main panel discussions. And two hands-on tutorials. Six international guest speakers from academic and industry are taking part in the CDMA2022 Conference. Also, the leading authority in the artificial intelligence (AI) field in Saudi Arabia will deliver the keynote speech at the conference. CDMA2022 Conference is organized at the premises of Prince Sultan University (PSU) in Riyadh, Saudi Arabia. PSU University is located at the center of Riyadh City with easy access to the city landmark locations.

CDMA2022 Conference will start officially on the first of March 2022. The hands-on tutorials are scheduled one day earlier to provide an opportunity to the conference attendees to enjoy all conference activities. The guest speeches, panel discussions and technical sessions will start on the first of March 2022. Top-quality papers are assigned to each technical session. The themes covered by these sessions include applications of machine learning, data science/analytics algorithms, deep learning models, intelligent storage and retrieval technologies, application of machine and deep learning in the areas of medicine, bioinformatics, natural language processing (NLP), remote sensing, Internet-of-things (IoT) networks, secure network systems, energy-efficient, green computing technologies, and quantum computing. Some accepted papers address benchmark datasets specific to Saudi Arabia including healthcare, Arabic NLP and Arabic sign language.

Building on the success of CDMA2020 Conference, only top-quality papers were accepted in this year's edition with an acceptance rate of 37%. To maintain the conference quality, tremendous efforts were exerted by the dedicated members of the conference technical committee (TPC). The TPC members were very meticulous in their review process while evaluating the submitted papers. Papers submitted to the CDMA2022 Conference were diversified geographically and technically (including academia, public and private industry sectors). All accepted papers will appear in the IEEE Xplore electronic database.

Finally, on behalf of the CDMA2022 Organizing Committees, we extend our invitation to all of you to join us at the CDMA2022 Conference from 28th of February 2022 to 3rd of March 2022 in PSU University, Riyadh, Saudi Arabia. We sincerely hope that you enjoy the conference program and activities.

II. KEYNOTE SPEAKERS

A. Prof. Majid Altwaijri, Chief Executive Officer (CEO) at the National Center for AI (NCAI) and CEO of Strategy Management Office at Saudi Data & AI Authority (SDAIA)

Keynote speaker biography: Prof. Majid Altwaijri is the Chief Executive Officer (CEO) at the National Center for AI (NCAI) and CEO of Strategy Management Office at Saudi Data & AI Authority (SDAIA). He has worked for 7 years

as the Managing Director of Health & Public Services in the Middle East, North Africa & Turkey at Accenture. He spent 9 years at King Saud bin Abdulaziz University for Health Sciences as a Professor and vice-president (VP) for Technology and a board member. He has been the Chief Information Officer (CIO) at the National Guard Health Affairs for 16 years. Dr. Altuwaijri is the Co-Founder of the Saudi Association for Health Informatics (SAHI), established in 2005. He has been appointed as the SAHI President for two terms. Prof. Altuwaijri has received his Master of Science (M.S.) and Doctor of Philosophy (Ph.D.) degrees from the University of Louisiana at Lafayette, USA (1990-1995). He holds a MCPM Master's Certificate in Project Management from the University of Houston-Clear Lake, USA since 2006. In 2012, he enrolled in the General Management Program (GMP) at the Harvard Business School (HBS), Harvard University, USA.

B. Prof. Gilbert Strang, Professor of Mathematics, Massachusetts Institute of Technology (USA)

Talk title: Linear Algebra and Deep Learning: Connections and Insights

Talk overview: Linear algebra has joined calculus as an essential subject in understanding modern science and engineering. Universities need to teach the high points of the subject in a constructive way: not proofs but ideas and useful factorizations. Those include the classical applications of eigenvalues and the modern applications of singular values in data science. Deep learning creates matrix weights from the training data. Those weights are chosen so that a sequence of steps from the input layer to the output will model the data. Each step is a matrix multiplication followed by a simple nonlinear operation like ReLU: keep the positive part and zero out the negative part of each layer. The resulting model has been highly successful when applied to new data. Understanding this success is essential to improving and accelerating deep learning.

We will identify key ideas in linear algebra that are part of the understanding.

Keynote speaker biography: Prof. Gil Strang is a professor of Mathematics at the Massachusetts Institute of Technology (MIT), USA. He has published 12 mathematics textbooks and recording his MIT lectures for Math18.06 -- 18.065 - 18.085 on the MIT OpenCourseWare (ocw.mit.edu). Those lectures have had more than twelve million viewers (also on youtube/mitocw). His lectures are freely available through MIT OpenCourseWare.

C. Prof. Yaser Abu-Mostafa, Professor of Electrical Engineering and Computer Science at the California Institute of Technology, USA

Talk title: AI and Machine Learning: The role of research and education

Talk overview: Artificial intelligence (AI) and Machine Learning have evolved into a mature technology that touches on every aspect of our life. Every country and every corporation must keep up with this technology to maintain its competitive advantage. To do this, what are the roles of research, development, and education? What is the right mix and the right emphasis that would enable us to make the most of the AI revolution?

This talk will address these questions based on my experience as a researcher, educator, AI practitioner, and scientific advisor.

Keynote speaker biography: Prof. Yaser Abu-Mostafa is Professor of Electrical Engineering and Computer Science at the California Institute of Technology, USA, Chairman of Paraconic Technologies Ltd, and Chairman of Machine Learning Consultants LLC. He is known for his research and educational activities in the area of machine learning. Prof. Abu-Mostafa studied at Cairo University, where he earned the BSc degree in 1979. He earned the MS degree at the Georgia Institute of Technology in 1981 and the PhD degree at the California Institute of Technology in 1983. He has been on the faculty of the California Institute of Technology since 1983. In 1987, Prof. Abu-Mostafa cofounded the Conference on Neural Information Processing Systems (NIPS), a major machine learning meeting. He is known for his recent textbook on machine learning. He has also developed an online course about machine learning.

D. Prof. Nizar Habash, Professor and Program Head of Computer Science at New York University Abu Dhabi (NYUAD), UAE

Talk title: Arabic Natural Language Processing: Challenges, Solutions and Outlook

Talk overview: Natural language processing, from document search to machine translation and question answering, has become an important part of our daily lives. The quality of such technologies for Arabic continues to lag behind those for English and other languages. In this talk, we present the main challenges to Arabic NLP. First, Arabic is a complex language – it has a high degree of orthographic ambiguity and variety, morphological richness, and dialectal variants. Secondly, research and development for Arabic language technologies also suffer from limited human and data resources, as well as collaborations. We present several efforts addressing these challenges situated in a bigger vision for the future of Arabic and Artificial Intelligence.

Keynote speaker biography: Prof. Nizar Habash is a Professor of Computer Science at New York University Abu

Dhabi (NYUAD). He is also the director of the Computational Approaches to Modeling Language (CAMEL) Lab. Prof. Habash specializes in natural language processing and computational linguistics. Before joining NYUAD in 2014, he was a research scientist at Columbia University's Center for Computational Learning Systems. He received his PhD in Computer Science from the University of Maryland College Park in 2003. He has two bachelor's degrees, one in Computer Engineering and one in Linguistics and Languages. His research includes extensive work on machine translation, morphological analysis, and computational modeling of Arabic and its dialects. Prof. Habash has been a principal investigator or co-investigator on over 25 research grants. And he has over 200 publications including a book entitled "Introduction to Arabic Natural Language Processing". His website is www.nizarhabash.com.

E. Prof. Hafiz Malik, Professor of Electrical and Computer Engineering (ECE) at the University of Michigan–Dearborn, USA

Talk title: Securing Biometric Systems against Deepfake and Shallowfake Attacks

Talk overview: Advances in artificial intelligence and multimedia (audio, video, and image) synthesis and generation technologies are the main driver behind exponential growth of deepfakes and shallowfakes. The deepfake technology manipulates audio and video of a real person saying or doing things they never said or did. Deepfake technology is weaponizing information and social media platforms are adding fuel to the fire by spreading it instantaneously. This talk will discuss trajectory of deepfake and shallowfake technologies and associated threats to national security, financial institutions, and smart devices. This talk will present a framework to debunk deepfake attacks in real-time. This talk will also present findings of our on-going research on detecting fake audios and videos.

Keynote speaker biography: Hafiz Malik is Professor of Electrical and Computer Engineering (ECE) at the University of Michigan – Dearborn. Leading national and international sponsors have funded his research in the areas of automotive cybersecurity, cyber-physical system security, deepfake detection, multimedia forensics, steganography/steganalysis, information hiding, pattern recognition, and information fusion. He has published over 100 articles in leading peer-reviewed journals, conferences, and workshops. He is a founding member and chief operating officer (COO) of the Global Foundation for Cyber Studies and Research, a founding member of the Cybersecurity Center for Research, Education, and Outreach at UM-Dearborn and member of leadership circle for the Dearborn Artificial Intelligence Research Center at UM-Dearborn. He is also a member of the Scientific and

Industrial Advisory Board (SIAB) of the National Center of Cyber Security Pakistan. He is a member of MCity Working Group on Cybersecurity, since 2015.

F. Dr. Hassan Sawaf, Founder of aiXplain, San Francisco Bay Area, USA

Talk title: Authority in Artificial Intelligence (AI) Science

Talk overview: How can we create a healthy and responsible vision for artificial intelligence (AI) and prevent it from becoming monopolized? How do we create a diverse playing field to let people and organizations come together to exchange AI while still owning their data and decentralizing the authority of AI? What are the benefits of doing this and how can we handle AI in the most responsible way for our future?

Keynote speaker biography: Hassan Sawaf has 25+ years of experience in employing cutting-edge technologies for mission-critical business operations. He founded several machine learning organizations in small and large tech companies including AppTek, eBay, Amazon, and Facebook. Hassan has always had a passion for entrepreneurial endeavors, learning, building and sharing his knowledge with others since the genesis of his career. In the University, he had a role as a Senior Researcher at RWTH Aachen, researching in MT and Speech Recognition, statistical parsing and language modeling. Upon receiving his diploma, he started his first company in 2001 in Germany and sold it in 2004 to a US-based company, which brought him to the Washington, DC area. Hassan then moved on to become Chief Scientist and HLT Director at SAIC for 2 years. He was responsible for R&D and technology development in the SAIC HLT division, supporting the team with deeper understanding of HLT and the business surrounding the technology. When relocating to the San Francisco Bay Area, he joined eBay as Director of Machine Translation, Human Language Technology and Head of AI for several years, and transitioned to external roles as an Advisor, equity partner and member of the board of numerous start-up and smaller established tech companies in the past 5 years. Currently, as Founder of aiXplain, he's on a mission to deeply democratize AI. aiXplain will make AI accessible to businesses and individuals alike, and ensure equitable and fair benefit to everyone involved. Hassan is also a founding board member of the PI Campus School headquartered in Rome, Italy, and investor, board member, and/or advisor to a handful of promising startups and established companies in the US, Canada, Germany, and UAE among them Tarjama, AppTek, and Embodied. In his spare time, he likes to read up on research, experience speed through skiing, one-wheeling, car racing, and flying and he also likes to build vehicles and little houses. During the pandemic, he rediscovered his passion for camping, tenting, and watching the sunrise and set, as he did not travel as usual.

E. Dr. Yves Lostanlen, Chief Business Officer, aiXplain, San Francisco Bay Area, USA

Talk title: Artificial Intelligence (AI) in networks, improve robustness of the models to “open” the black box

Talk overview:

Mobile networks need to be resilient and reliable. AI has the promise to optimize network performance, but the black box nature of the machine learning (ML) and deep learning (DL) models creates challenges for operators looking to adopt these techniques due to the difficulty to explain how the model takes a decision and the risk/liability associated when the model fails. This talk discusses the shortcoming of interpretability and new approaches that have been tested to audit models and move towards a more Human - AI hybrid system to mitigate these challenges.

Tutorial speaker biography: Yves Lostanlen is a leading expert in innovation technology transfer with a proven track record in academic research and high-growth companies, having spearheaded original multi-partner business solutions and complex transactions with clients at the crossroads of several sectors. Prior to joining aiXplain, Yves’s 25+ years of experience entails serving as Managing Director and Head of EMEA at Element AI; the CEO of a North American subsidiary of a Global Fortune 100 company; advisor at Facebook, Google, Bell Labs, Orange, and AT&T; mentoring and founding several startups; and working with 100+ scientific research labs. With a motivating leadership and coaching style, Yves has also led multiple-site global teams, delivering transformative Enterprise Software platforms and services. Adding onto his expertise in the intersection of business and technology, Yves is an active contributor to the creation of business and innovation ecosystems, particularly in artificial intelligence and the internet of things, across Europe and North America. Yves holds a MEng and PhD in Electrical Engineering, a Doctorate of Science (Habilitation) in Computer Science and Physics, and an MBA from MIT Sloan School of Management. He is also an Adjunct Professor at the University of Toronto and Senior Executive Fellow at the University of Waterloo, Canada.

III. TUTORIALS

A. Dr. Issam Laradji, Research Scientist at ServiceNow Company, Vancouver, Canada

Tutorial title: Build End-to-End Machine Deep Projects and Manage Large-Scale Experiments

Tutorial Overview: In this tutorial I will briefly describe best practices for building large-scale reproducible deep Learning projects, and for defining, launching and visualizing a large set of experiments. On Google Colab, starting from an empty project, I will show how to compare between 4 optimizers on two datasets and obtain visualizations that are ready to be presented in a research

paper or project report. By the end of the tutorial, you will be able to build a deep learning benchmark, launch large-scale experiments, visualize them, debug failed experiments, and generate publishable results for a final machine learning report.

Tutorial speaker biography: Issam Laradji is currently a Research Scientist at ServiceNow company. He is interested at developing methods that require weaker supervision in order to reduce the amount of effort humans need for acquiring training sets. His current focus is on 3D computer vision that touch on rendering, synthesizing novel scenes, and learning to infer physical properties of objects. On the side, he occasionally works on new optimization methods for deep learning. He also continuously works on and maintains Haven-AI (<https://github.com/haven-ai/haven-ai>) which is a library for running, managing and visualizing large scale experiments in the hopes to make machine learning research accessible and reproducible to people.

B. Mr. Eyad Sibai, Chief Data Scientist at Al-Souda Development Authority

Tutorial title: Machine Learning Pipelines Using Scikit-learn and Pandas

Tutorial Overview: Machine learning pipeline consists of multiple steps to train a model covering data extraction, preprocessing, and model training. In this tutorial we will learn how to write clean, well-organized machine learning pipelines using pandas and scikit-learn. We will cover the design principles of scikit-learn, how to use pandas with scikit-learn, writing custom transformers, dealing with heterogeneous data (numerical, categorical, text, and dates), debugging your pipeline, and ways to deploy the model after the training.

Tutorial speaker biography: Mr. Eyad Sibai has over 12 years of seasoned experience in data science and artificial intelligence across multiple industries and diverse multicultural environments. With a proven track record in establishing, and building data and machine learning teams. Technical "hands-on" expertise in implementing and developing machine learning solutions and big data architectures. Determined, optimistic and engaging with strong diplomatic and interpersonal skills. Motivator and people leader focused on serving teams to enhance cohesion, ensure trust and reach performance excellence.

IV. COMMITTEES

The following presents the organization committee chairs and the technical program committee members for this year’s conference.

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